

Rendering Optimization

Problem

- One Mesh per voxel → one draw call per voxel
- Draw calls create CPU bottlenecks

Solution

- Group voxels by color
- Render with InstancedMesh
- One draw call renders many cubes

Trade-off

- Raycasting returns instanceId
- Need mapping: instanceId → voxel key

VoxelStore

- Voxel, (coordinate, colorId)
- Map<CoordKey, Voxel>
 - Fast lookup and updates

Render pipeline

VoxelStore



Voxel[]



GroupRender(colorId, voxels, indexToKey)

When generating the InstancedMesh, I iterate through the voxels in a deterministic order.
During this process, I push the voxel key into indexToKey using the same index that I use to create the instance.



InstancedMesh *with JSX*

Undo / Redo

Data layer

VoxelStatus

- coordinates[]
- colorId

Operation

- type
- timestamp
- prevStatus
- newStatus

Execution

New operation

- push to undo stack
- clear redo stack

Undo

- pop undo
- apply prevStatus
- push to redo

Redo

- pop redo
- apply newStatus
- push to undo

Load scene

- clear both stacks

DSL

- one operation can modify multiple voxels.